CSE4227 Digital Image Processing

Chapter 10 – **Image Segmentation (Part III)**

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Today's Contents

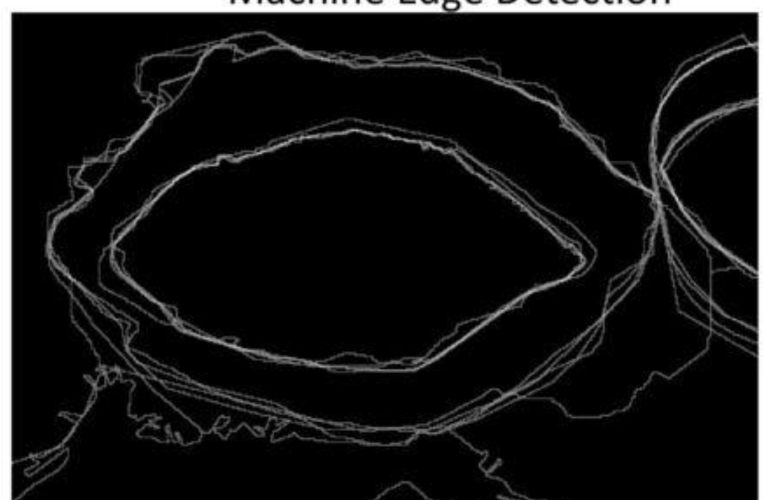
- Introduction to Hough transform
- Using gradient information to detect lines
- Representing a line
- The [a,b]-representation
- The [ρ,θ]-representation
- Algorithms for detection of lines

Chapter 10 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [**Section 10.2.7**]

Boundaries of Objects from Edges

Image

Machine Edge Detection



Human Boundary Marking

Knowledge about Boundary

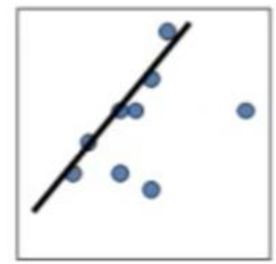
More

Knowledge

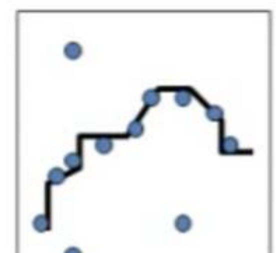
Less

- Can extract the complete closed contour of the object if its shape is known

- Can extract pieces of the boundary using general line or curve models



- Can just try to find a connected series of edge elements, using only heuristics on say, edge curvature

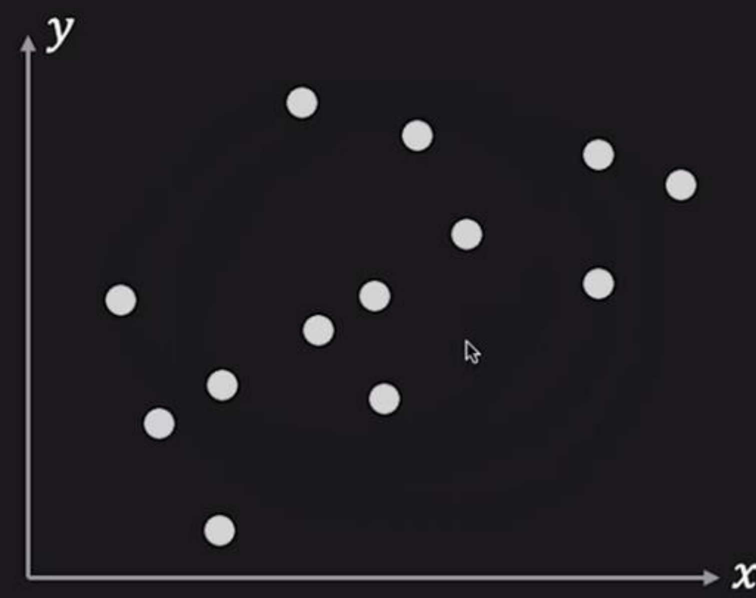


Hough Transform

- A way of finding edge points in an image that lie along a straight line or curve.
- Introduced by Paul Hough (1962)
- First goal is to find the location of lines in image.
- Given a set of edge points, find the line(s) which best explain the data.
- Relatively unaffected by gaps in lines, curves, and noise.
- Hough Transform is used to connect the disjointed edge points.

Hough Transform: Line Detection

Given: Edge Points (x_i, y_i)

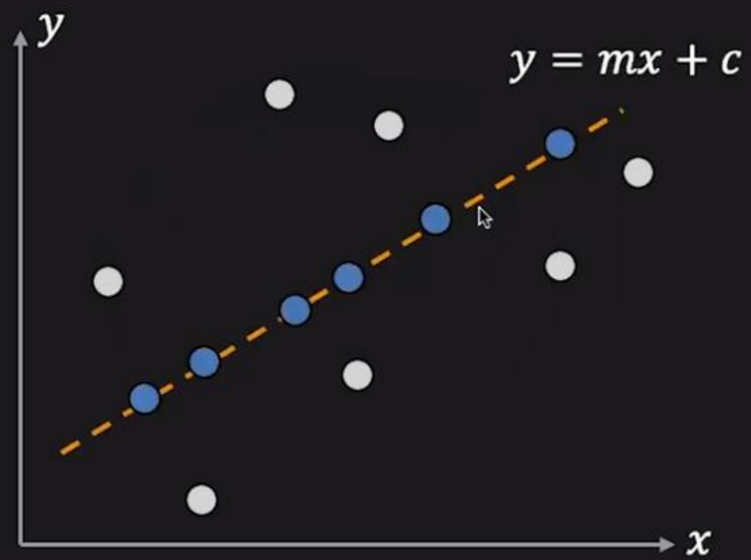


Hough Transform: Line Detection

Given: Edge Points (x_i, y_i)

Task: Detect line

$$y = mx + c$$



m = slop

c = intercept of the line

Hough Transform: Line Detection

Given: Edge Points (x_i, y_i)

Task: Detect line

$$y = mx + c$$

m = slop
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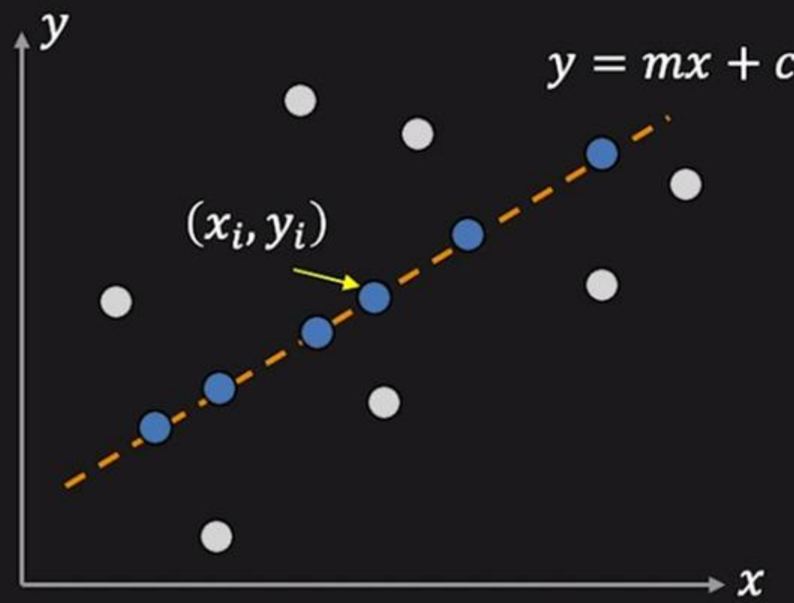
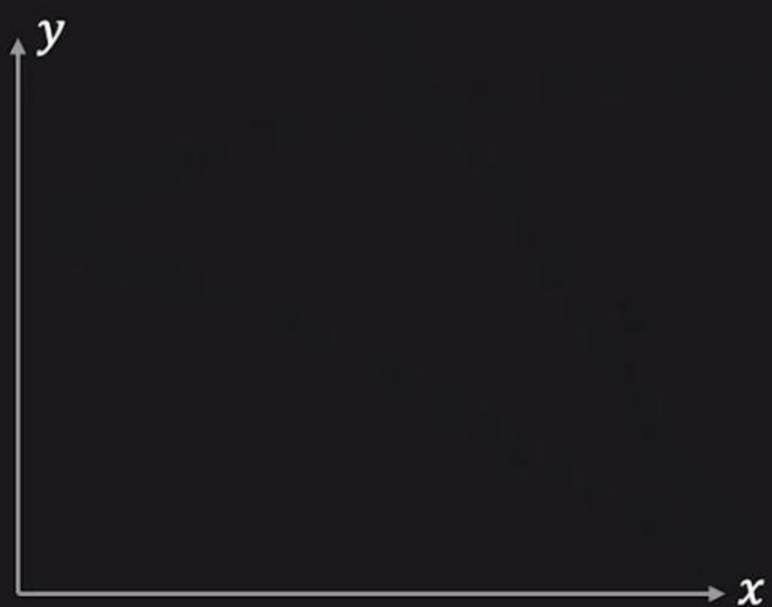


Consider point (x_i, y_i)

$$y_i = mx_i + c \quad \Longleftrightarrow \quad c = -mx_i + y_i$$

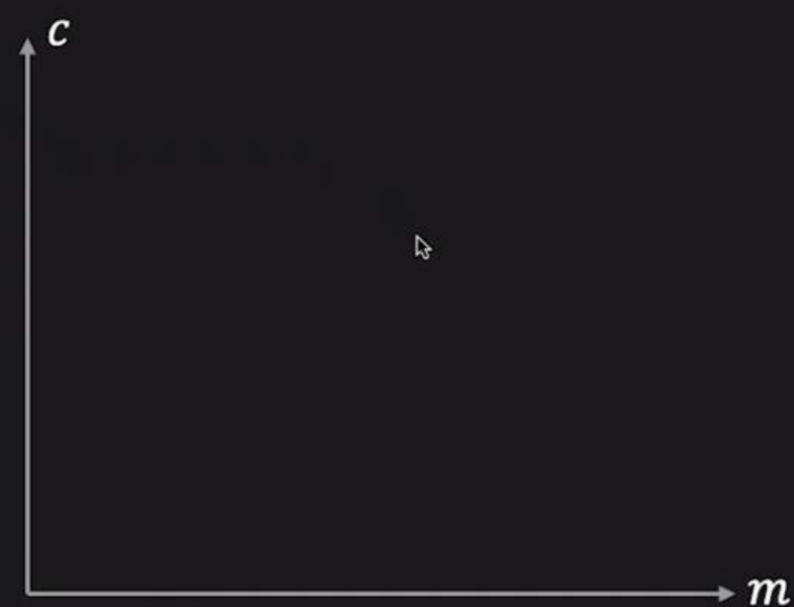
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

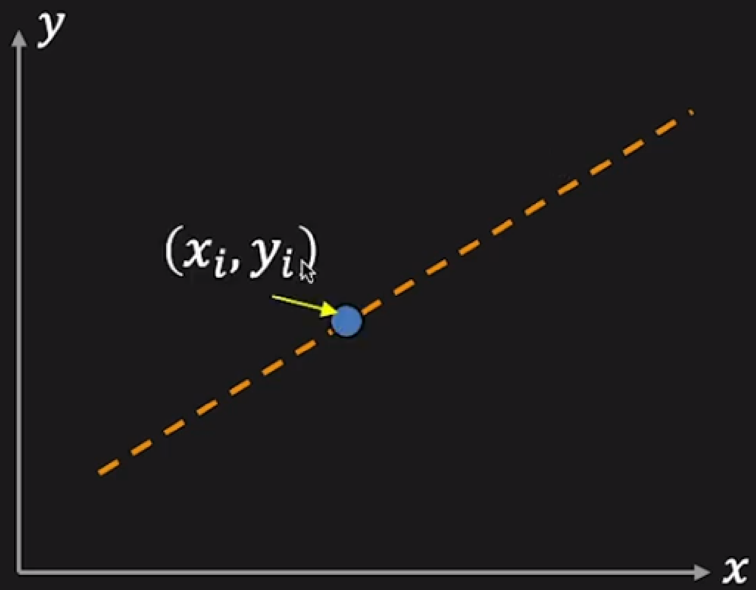
Parameter Space



$$c = -mx_i + y_i$$

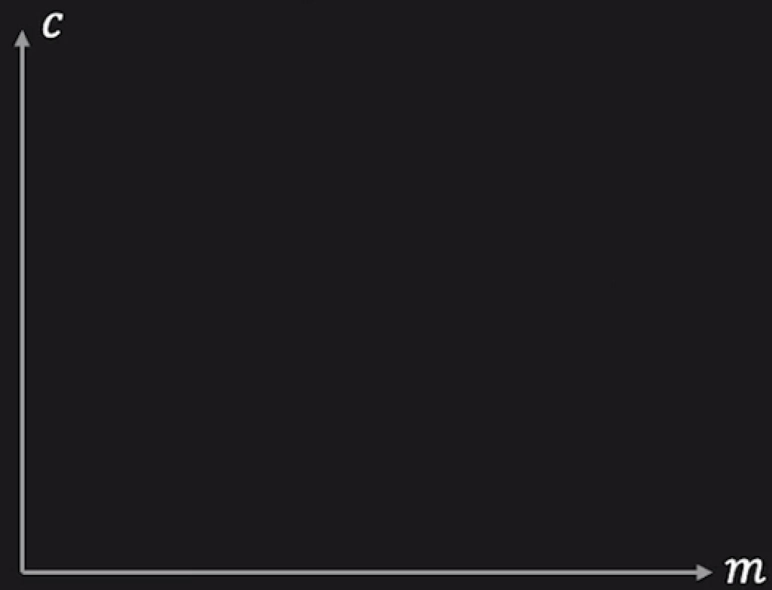
Hough Transform: Concept

Image Space



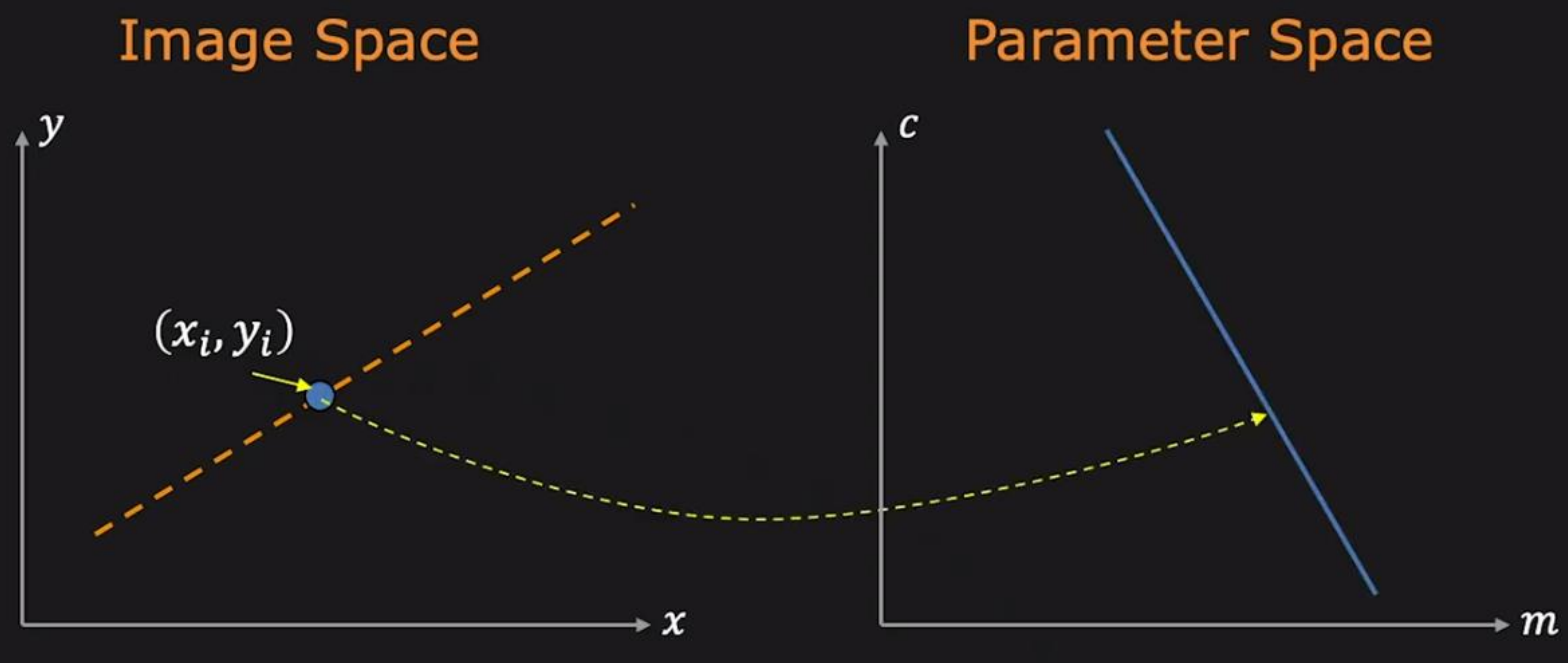
$$y_i = mx_i + c$$

Parameter Space



$$c = -mx_i + y_i$$

Hough Transform: Concept



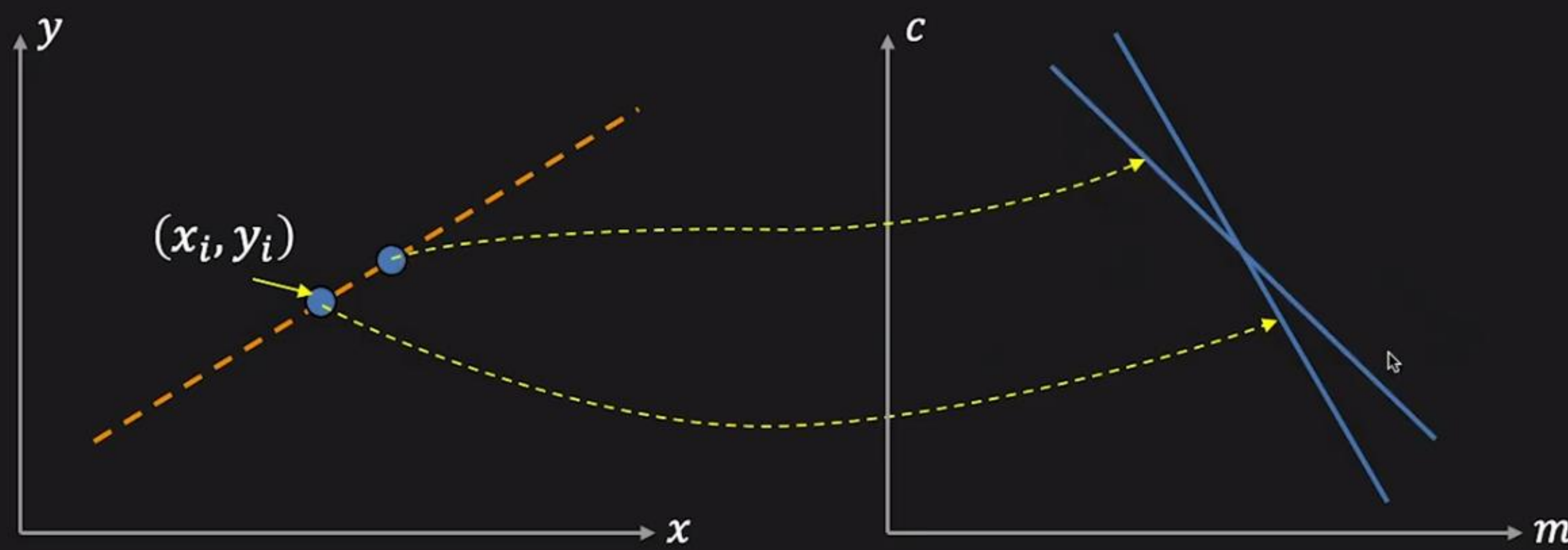
$y_i = mx_i + c$

$c = -mx_i + y_i$

Hough Transform: Concept

Image Space

Parameter Space



$$y_i = mx_i + c$$

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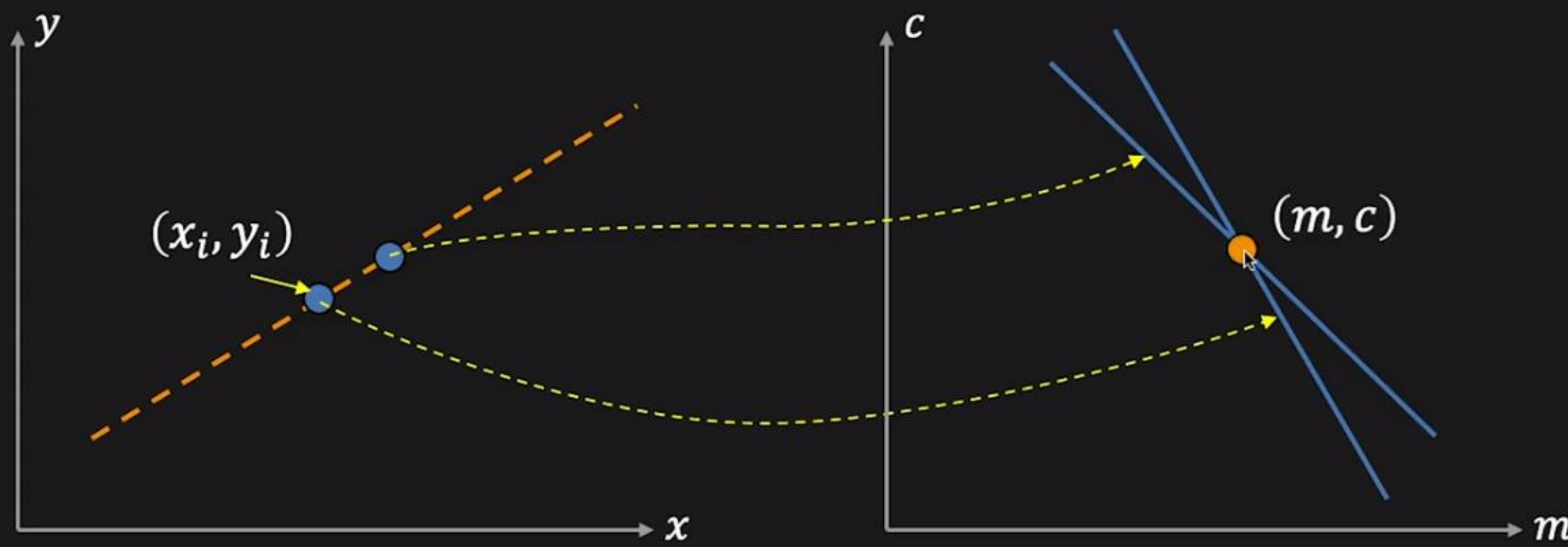
Hough Transform: Concept

Image Space

Parameter Space



$y_i = mx_i + c$

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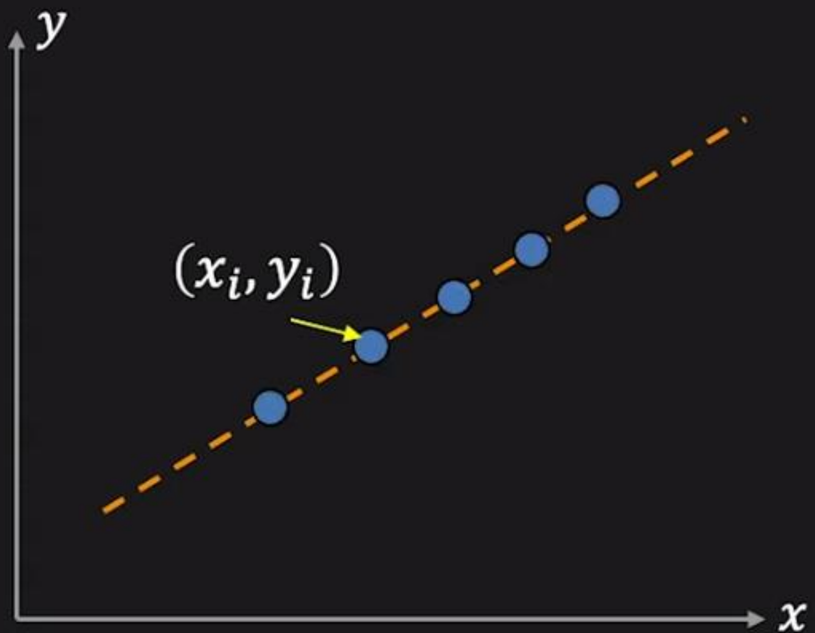
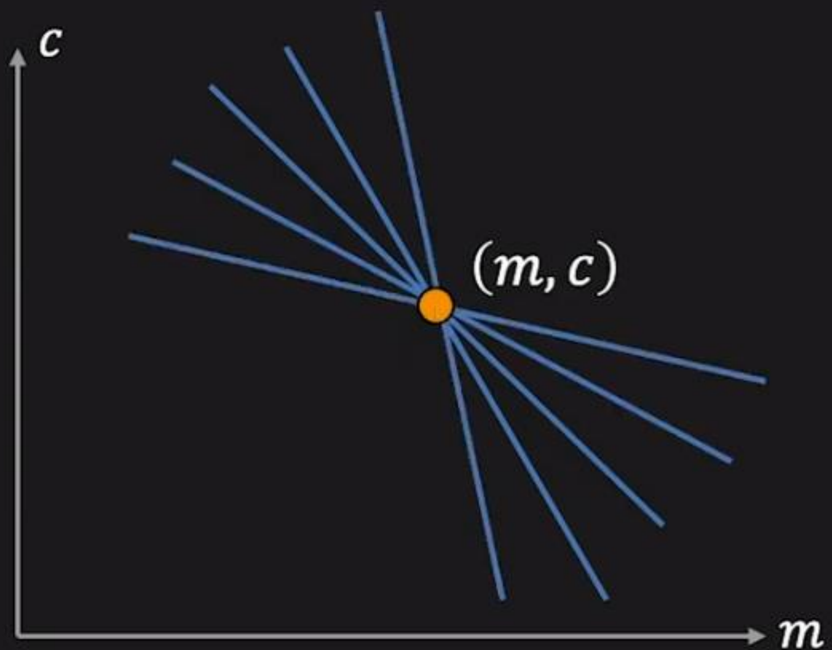
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

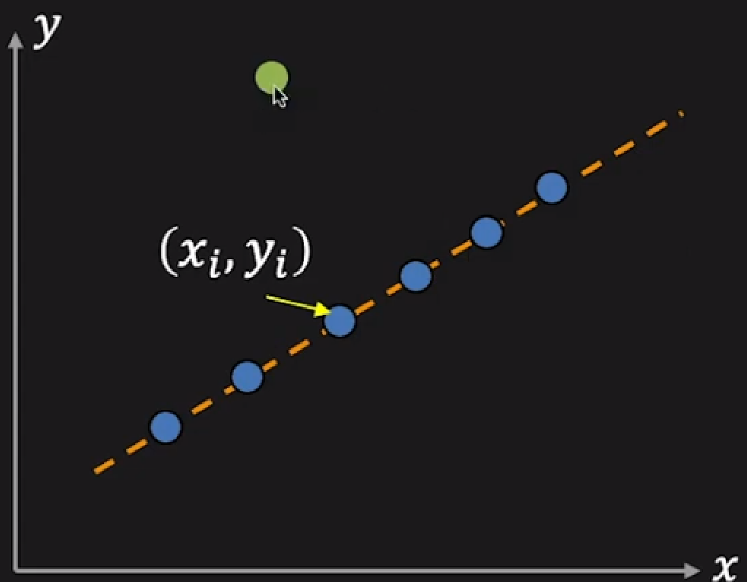
Parameter Space



$$c = -mx_i + y_i$$

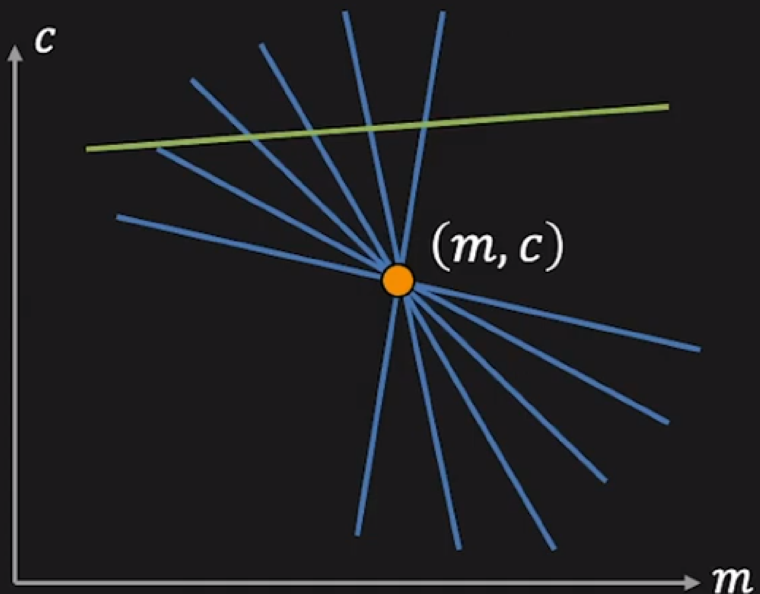
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

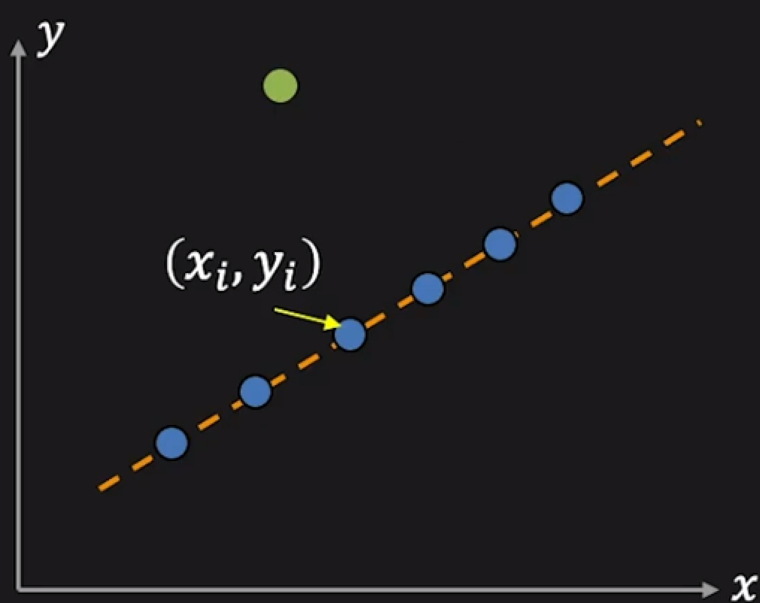
Parameter Space



$$c = -mx_i + y_i$$

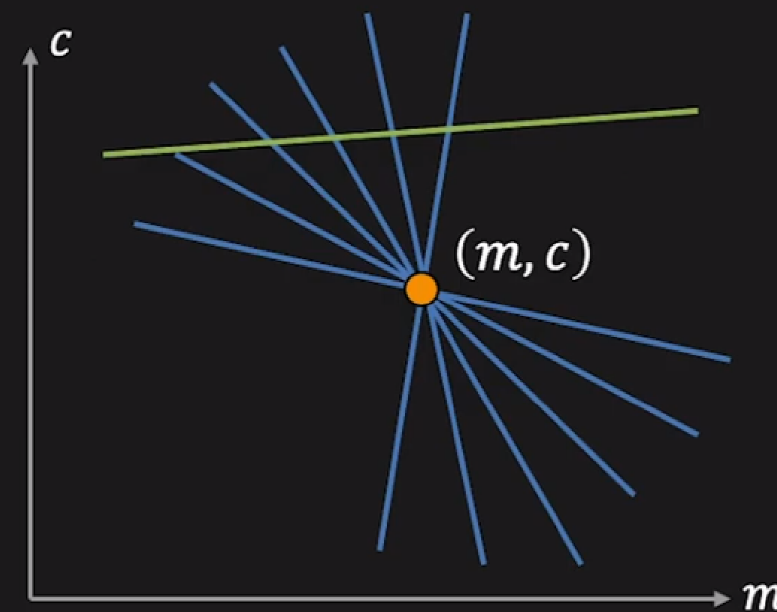
Hough Transform: Concept

Image Space



$$y_i = mx_i + c$$

Parameter Space



$$c = -mx_i + y_i$$

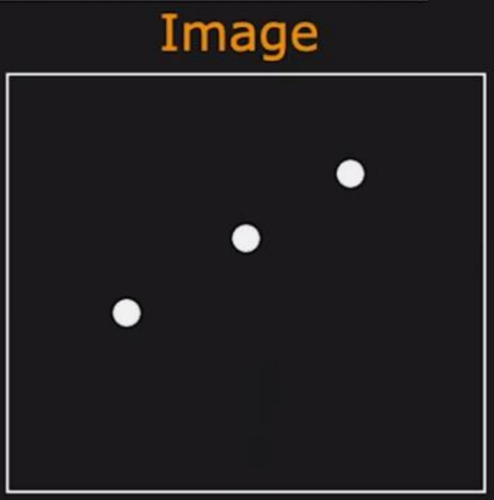
Point ⟷ Line

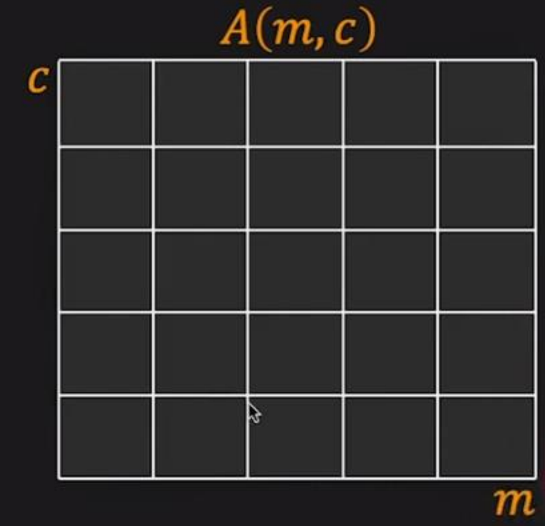
Line ⟷ Point

Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create accumulator array $A(m, c)$





Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create accumulator array $A(m, c)$

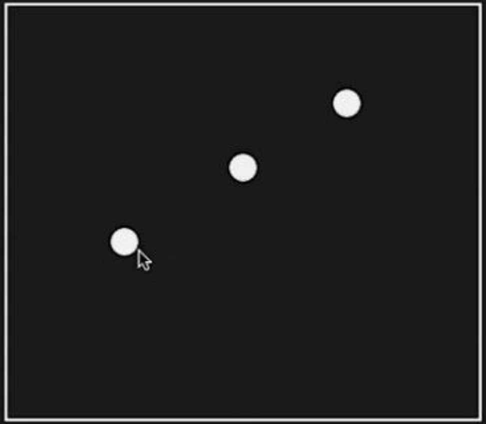
Step 3. Set $A(m, c) = 0$ for all (m, c)

Step 4. For each edge point (x_i, y_i),

$$A(m, c) = A(m, c) + 1$$

if (m, c) lies on the line: $c = -mx_i + y_i$

Image



$A(m, c)$

c					
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0
	0	0	0	0	0

Line Detection Algorithm

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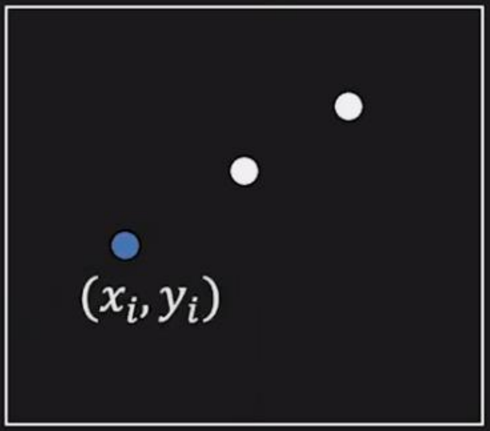

$A(m, c)$

c	1	0	0	0	0
	0	1	0	0	0
	0	0	1	0	0
	0	0	0	1	0
	0	0	0	0	1

Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create accumulator array $A(m, c)$

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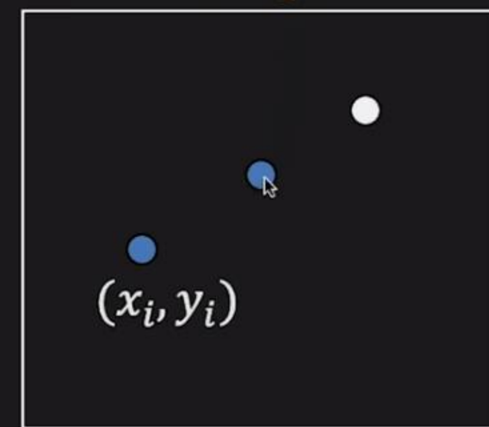
Image



$A(m, c)$

c					
	1	0	0	0	1
	0	1	0	1	0
	0	0	2	0	0
	0	1	0	1	0
	1	0	0	0	1

Line Detection Algorithm

Step 1. Quantize parameter space (m, c)

Step 2. Create accumulator array $A(m, c)$

Step 3. Set $A(m, c) = 0$ for all (m, c)

Step 4. For each edge point (x_i, y_i),

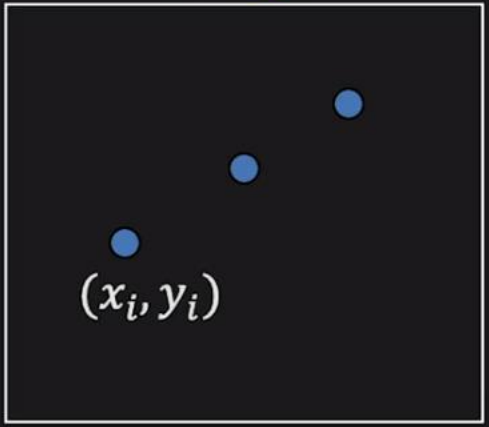
$$A(m, c) = A(m, c) + 1$$

if (m, c) lies on the line: $c = -mx_i + y_i$

Step 5. Find local maxima in $A(m, c)$

Draw the line using local maxima

Image



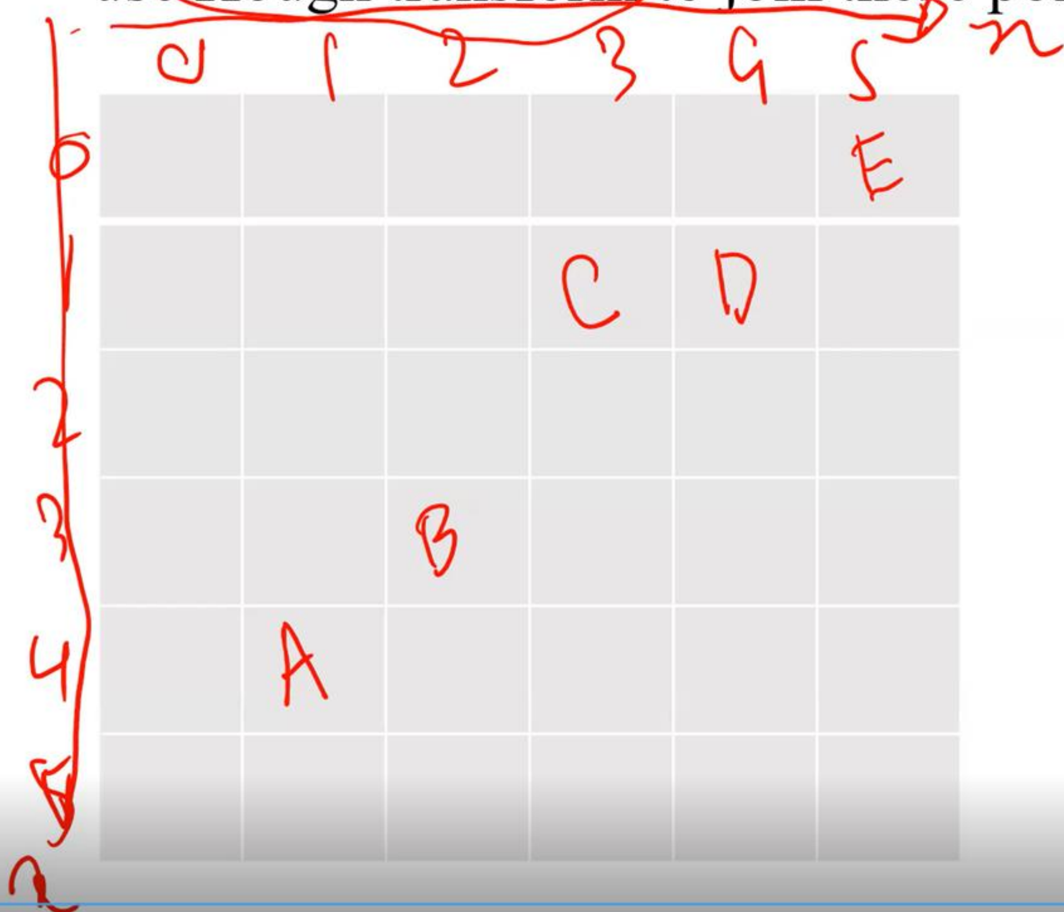
$A(m, c)$

c

1	0	0	0	1
0	1	0	1	0
1	1	$\boxed{3}$	1	1
0	1	0	1	0
1	0	0	0	1

Hough Transform Example

❖Given a set of points A(1,4), B(2,3), C(3,1) D(4,1) and E(5,0), use Hough transform to join these points.



$$y = mx + c$$

$$c = -mx + y$$

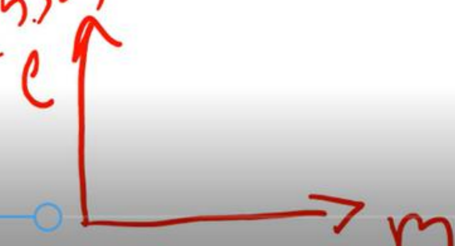
A(1,4) $c = -m + 4$

B(2,3) $c = -2m + 3$

C(3,1) $c = -3m + 1$

D(4,1) $c = -4m + 1$

E(5,0) $c = -5m + 0$



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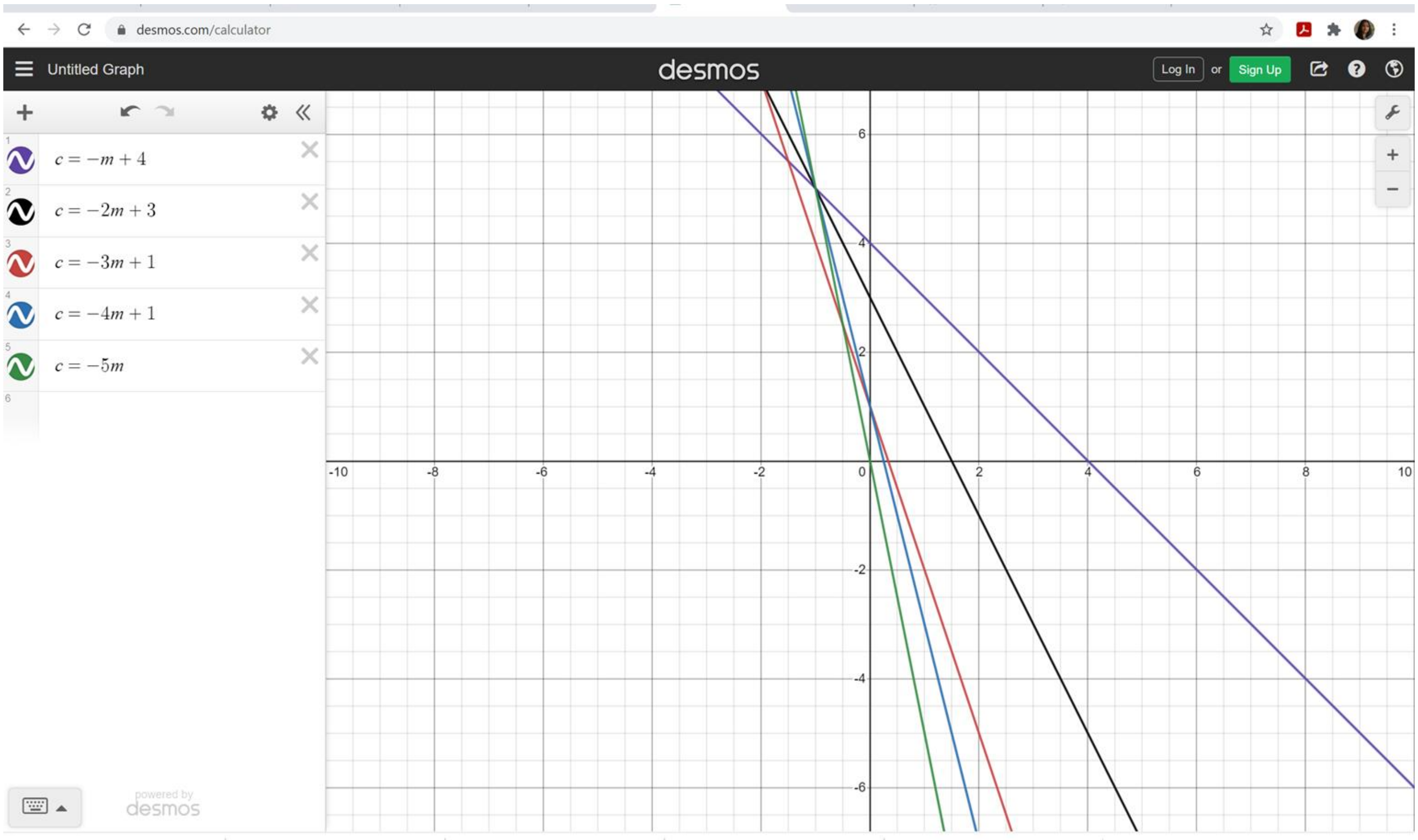


Hough Transform Example

- Given a set of points A(1,4), B(2,3), C(3,1) D(4,1) and E(5,0), use Hough transform to join these points.

https://youtu.be/Lbu9SnoucW4

Hough Transform Example

- Using Hough transform, show that the points (1,2), (2,2) and (3,4) are collinear. Also find the equation of the line.

What is the problem in (m,b) parameter space / Hough space algorithm

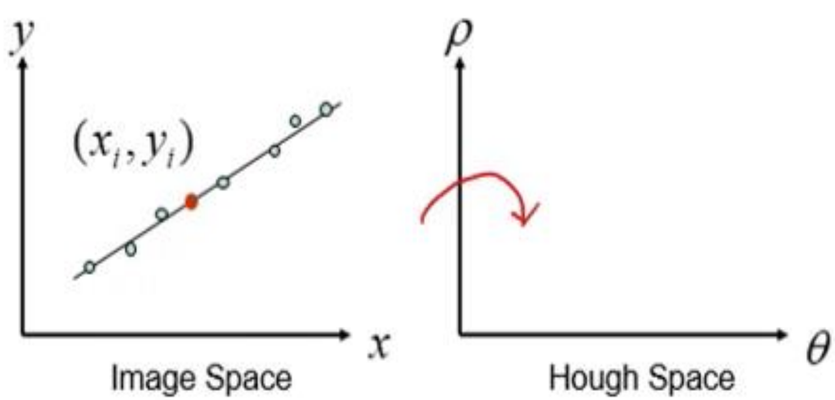
- Limitations of this algorithm is that it does not work for vertical lines i.e. lines those are parallel to the y-axis.
- Because they have undefined or infinity slope.
- Therefore these lines must be converted into polar coordinates.

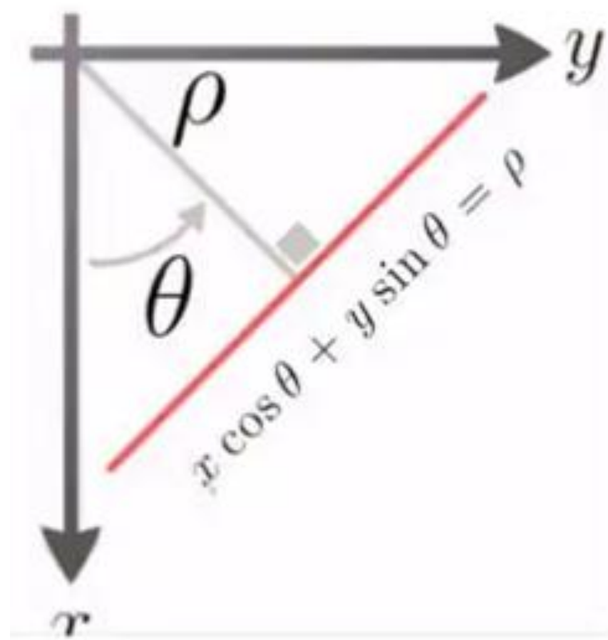
- Limitation of **previous algorithm** is that it does not work for vertical lines because they have infinity slope
- Therefore this **line** must be converted into **polar coordinates**

Equation of Line in polar:

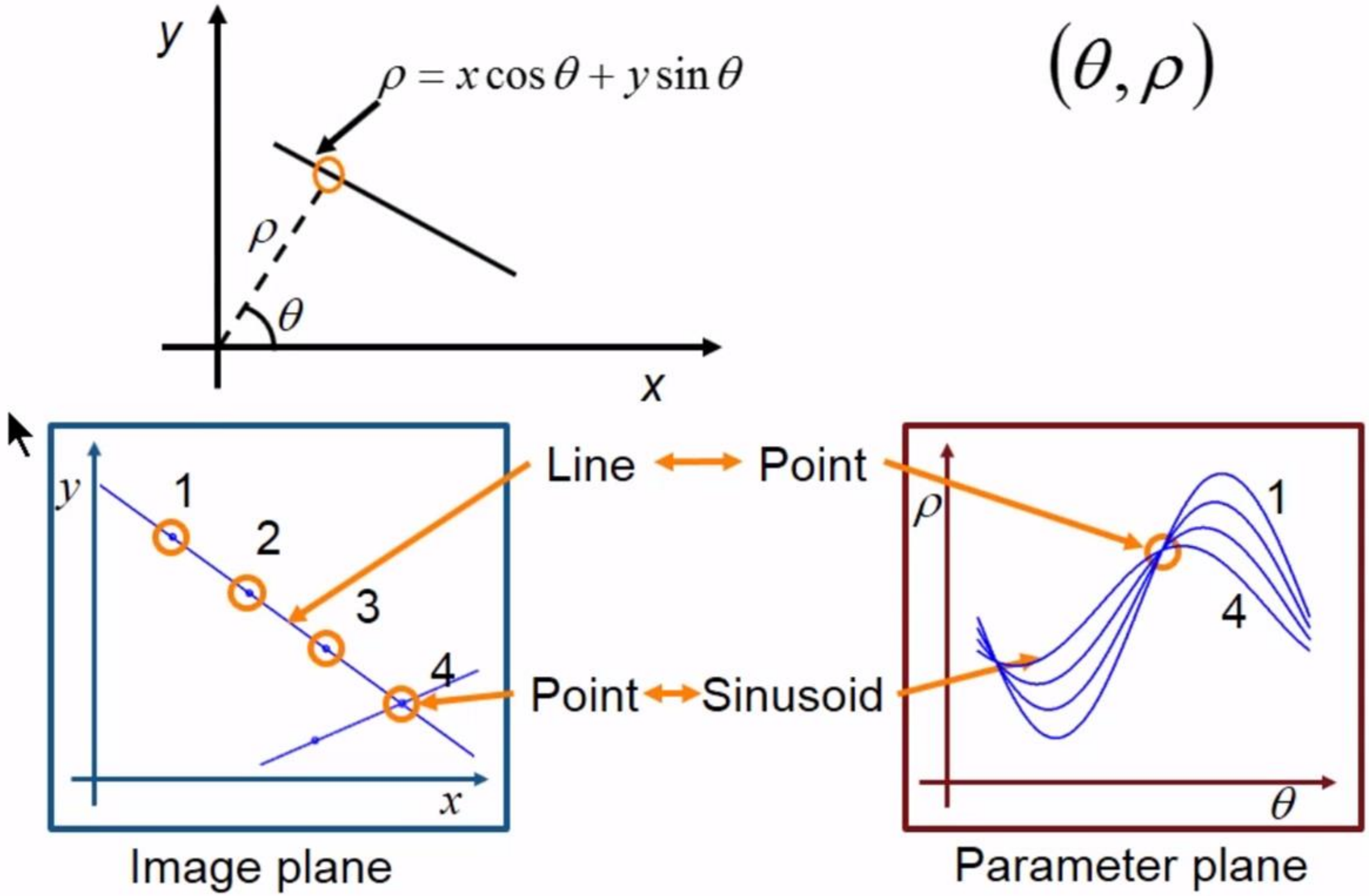
$$\rho = x\cos\theta + y\sin\theta$$

Where,
ϴ = angle between the line
ϱ = diameter



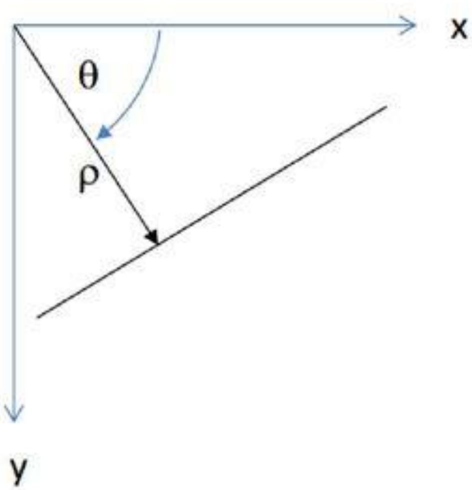


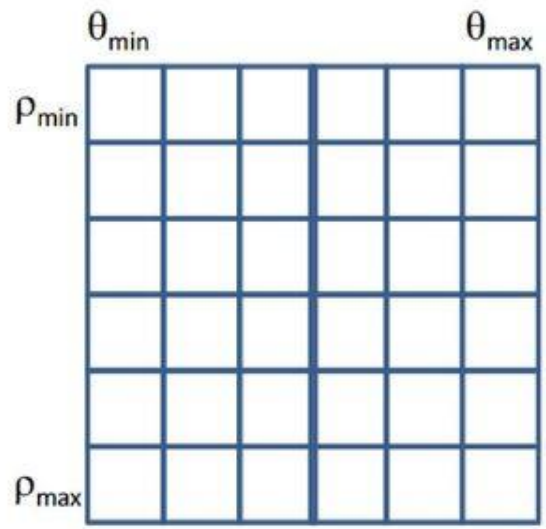
Standard Hough Transform



Polar Coordinate Representation of Line

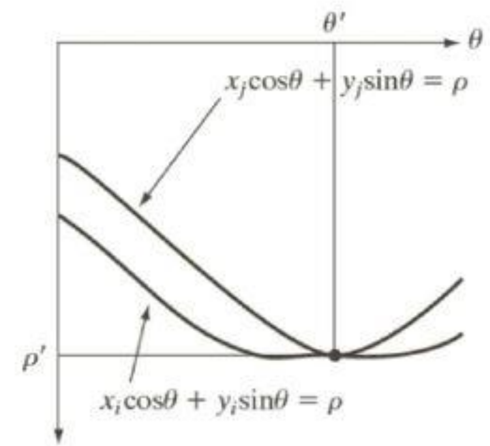
- $\rho = x \cos \theta + y \sin \theta$
  - Avoids infinite slope
  - Constant resolution





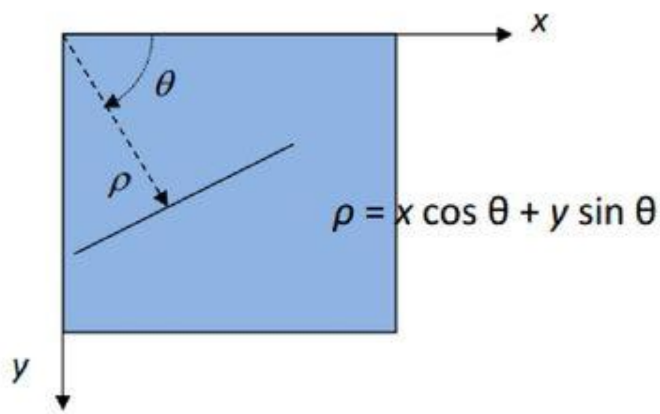
A(ρ,θ)

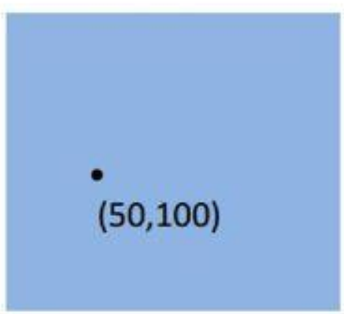
The parameter space transform of a point is a sinusoidal curve

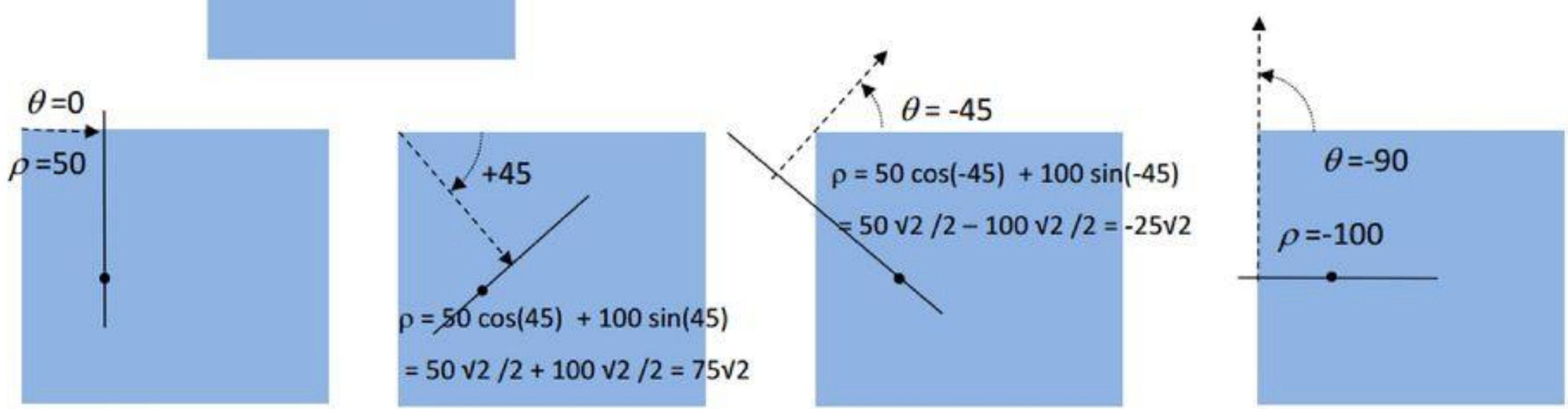


Hough Transform

- Angle, axis conventions
  - angle range is -90°..+89°
  - rho range is –dmax..+dmax
    - dmax is the largest possible distance
- Example of a point at (x,y) = (50,100)







Example

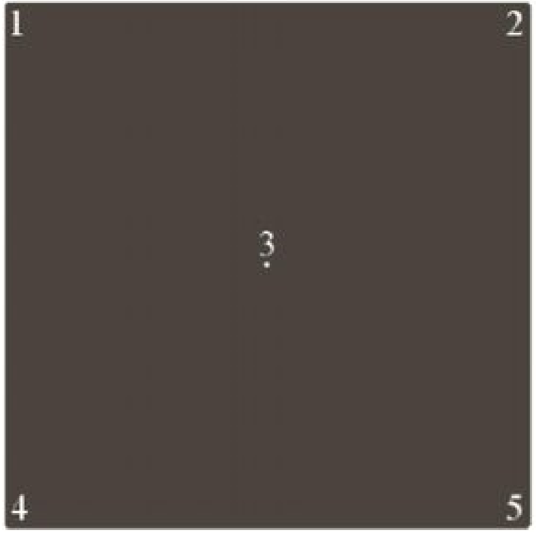
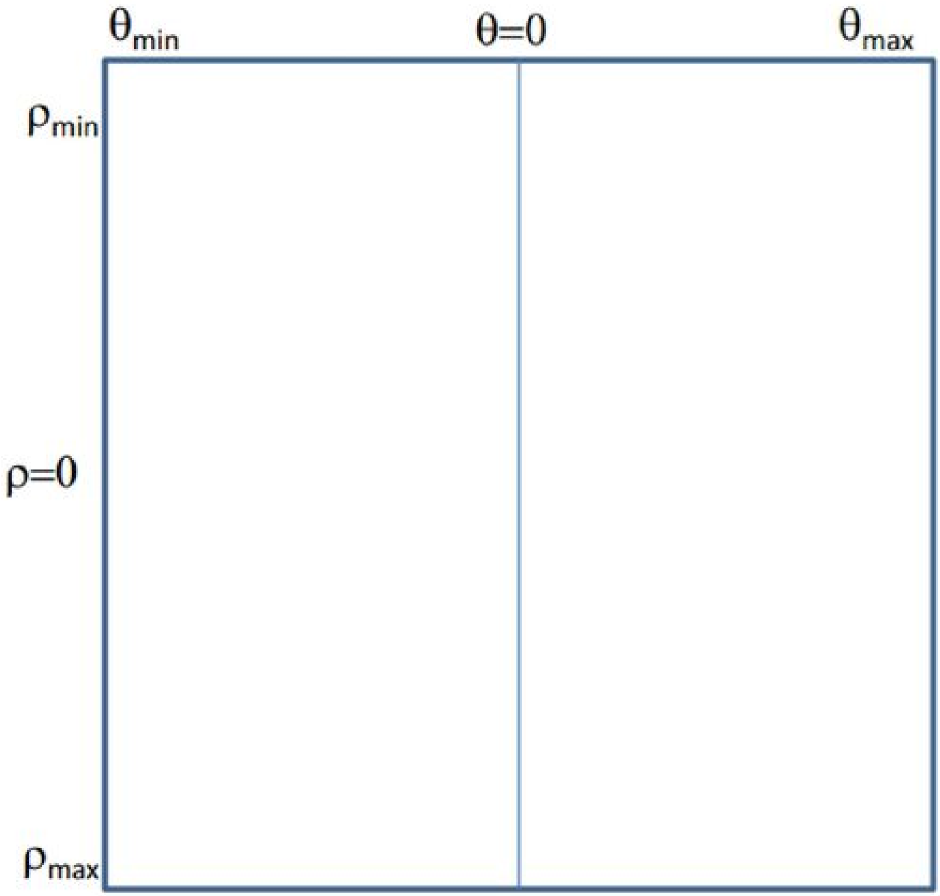
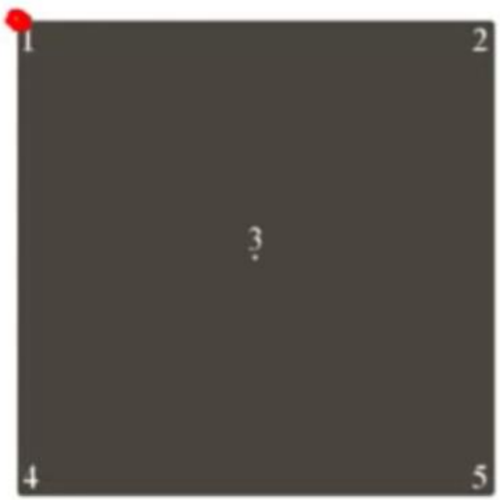


Image of size 100x100, containing 5 points (at the corners)



https://youtu.be/kMK8DjdGtZo

Example



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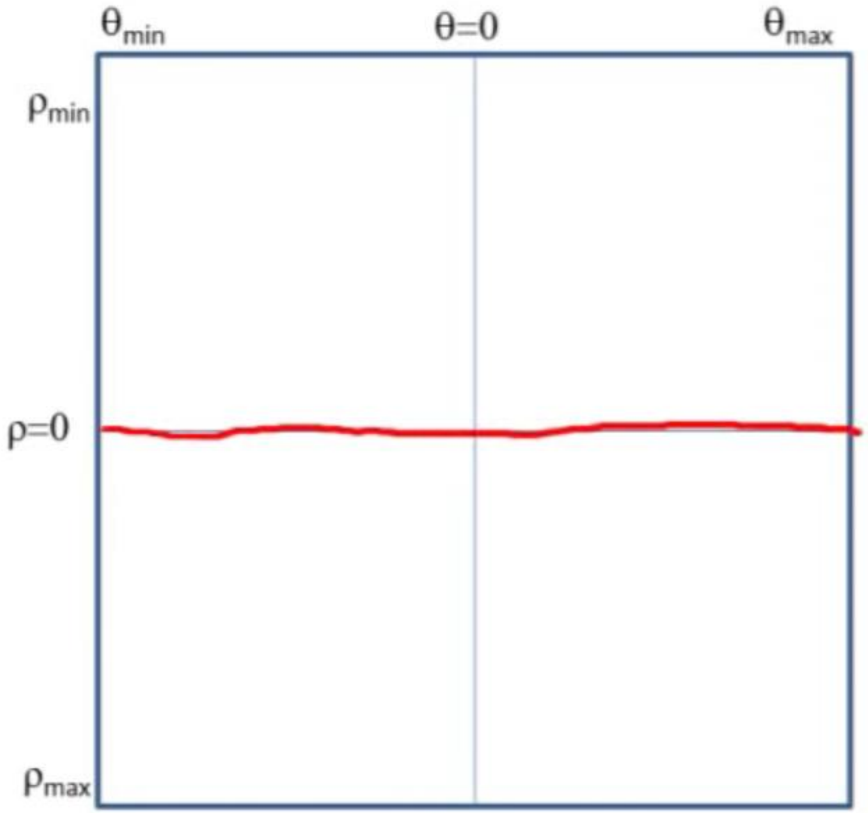


Example

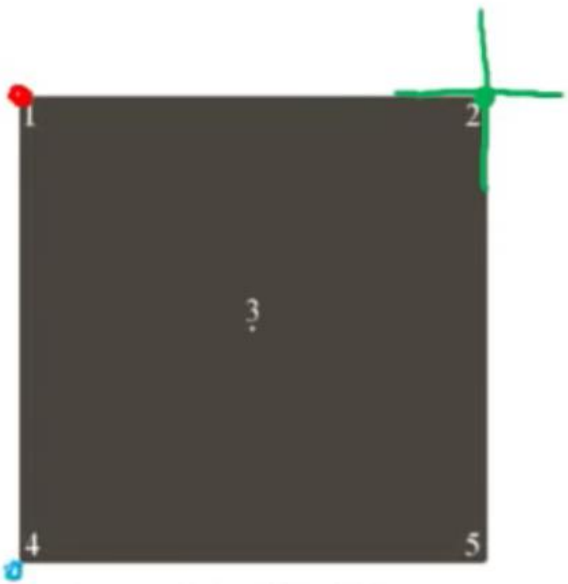
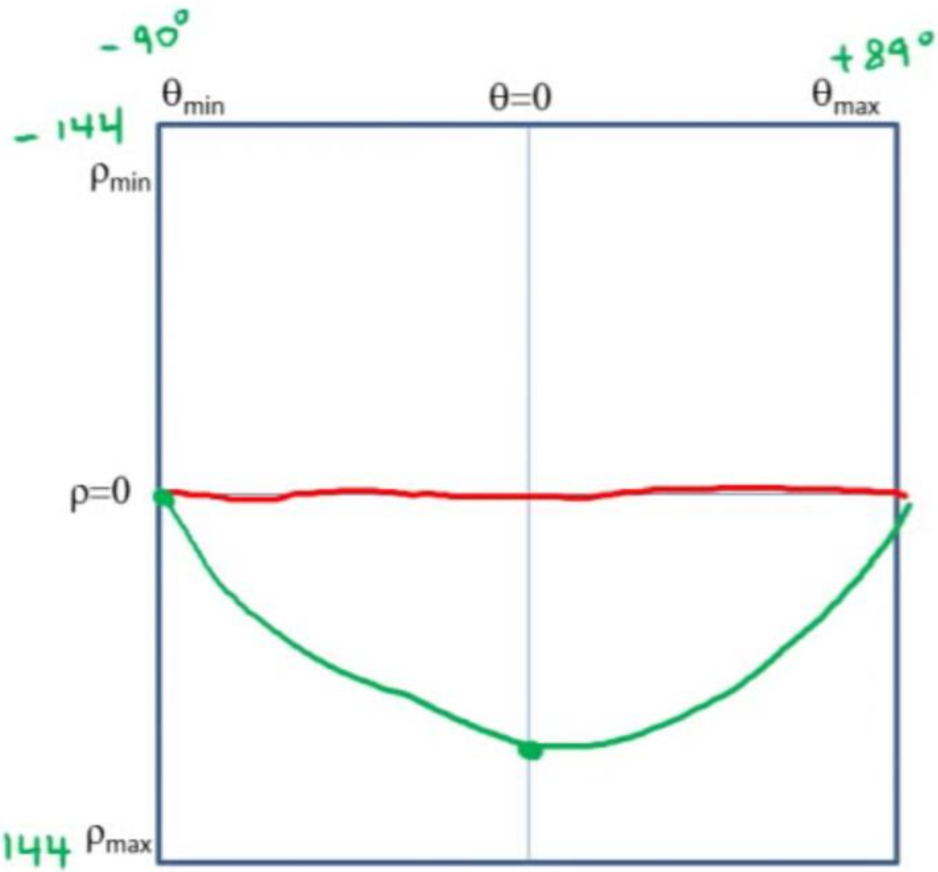
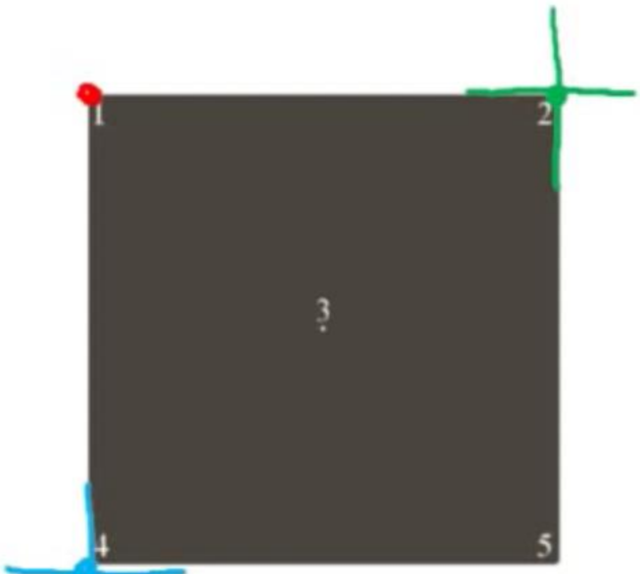


Image of size 100x100, containing 5 points (at the corners)



Example



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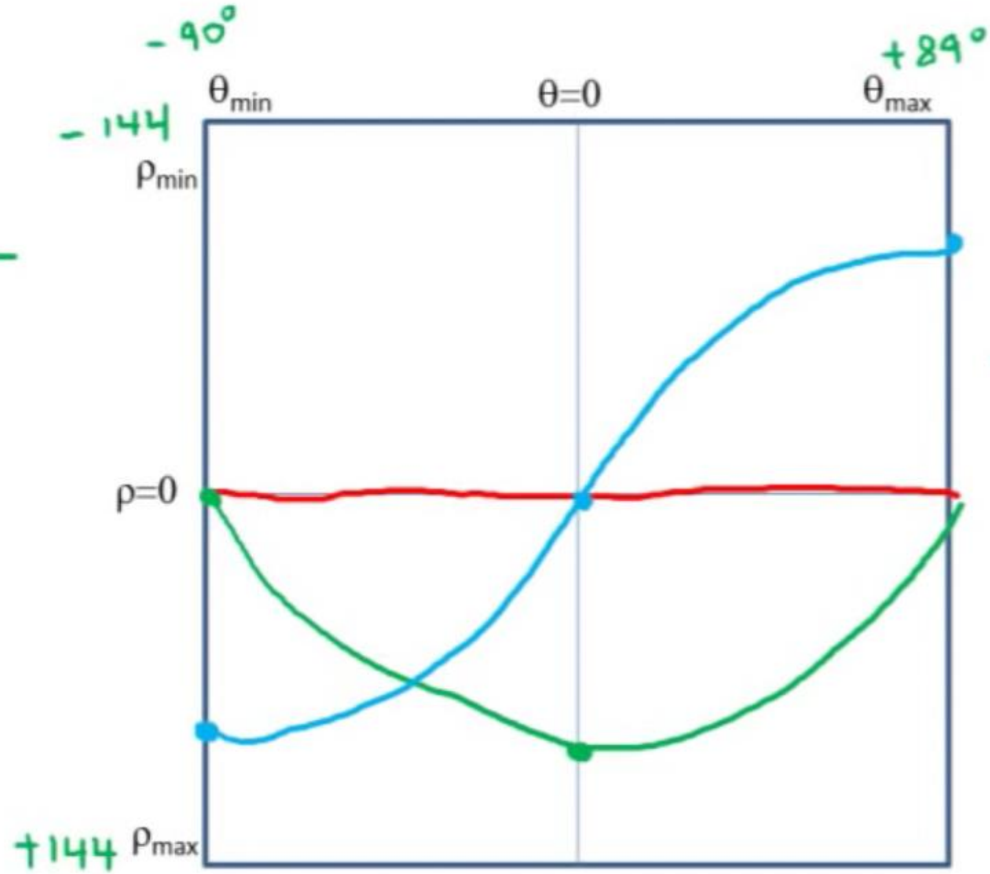
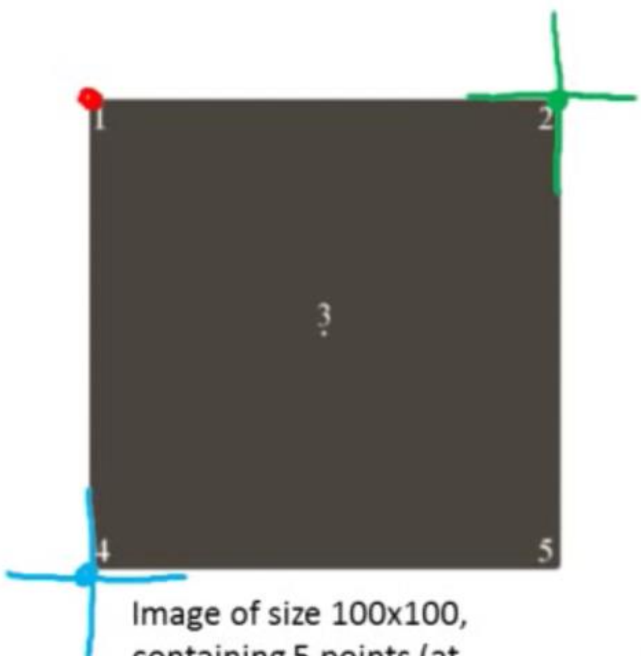


Example



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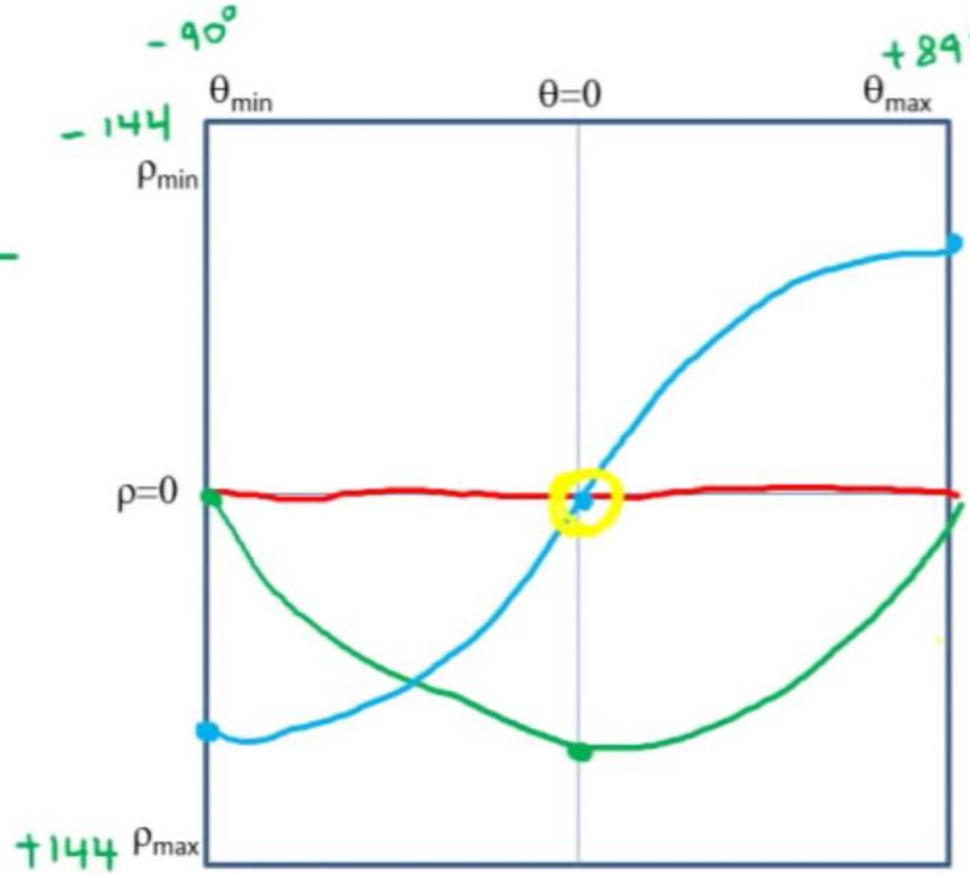


Example

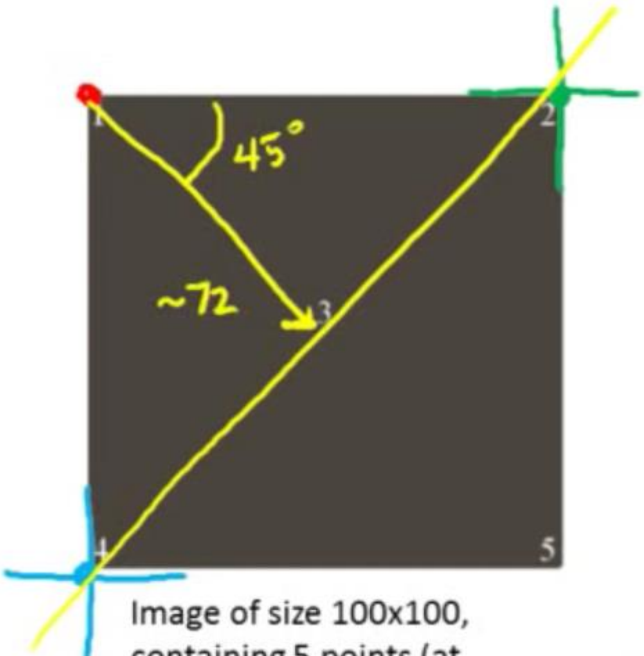
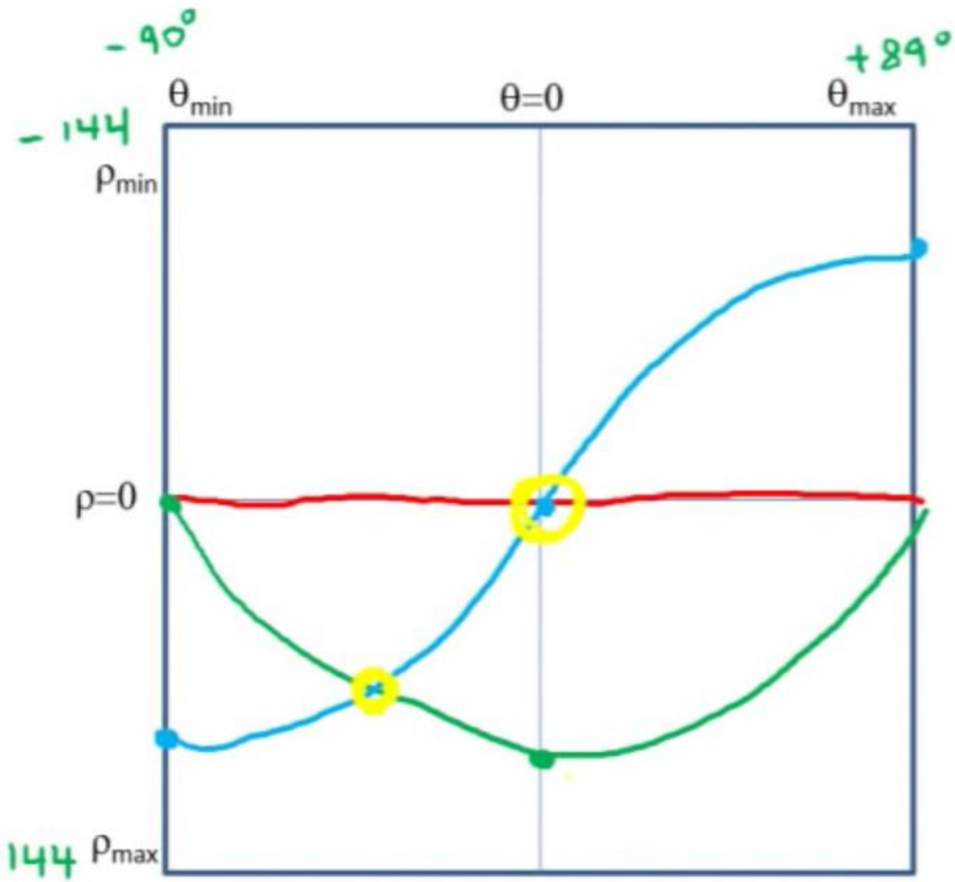


Image of size 100x100, containing 5 points (at the corners)

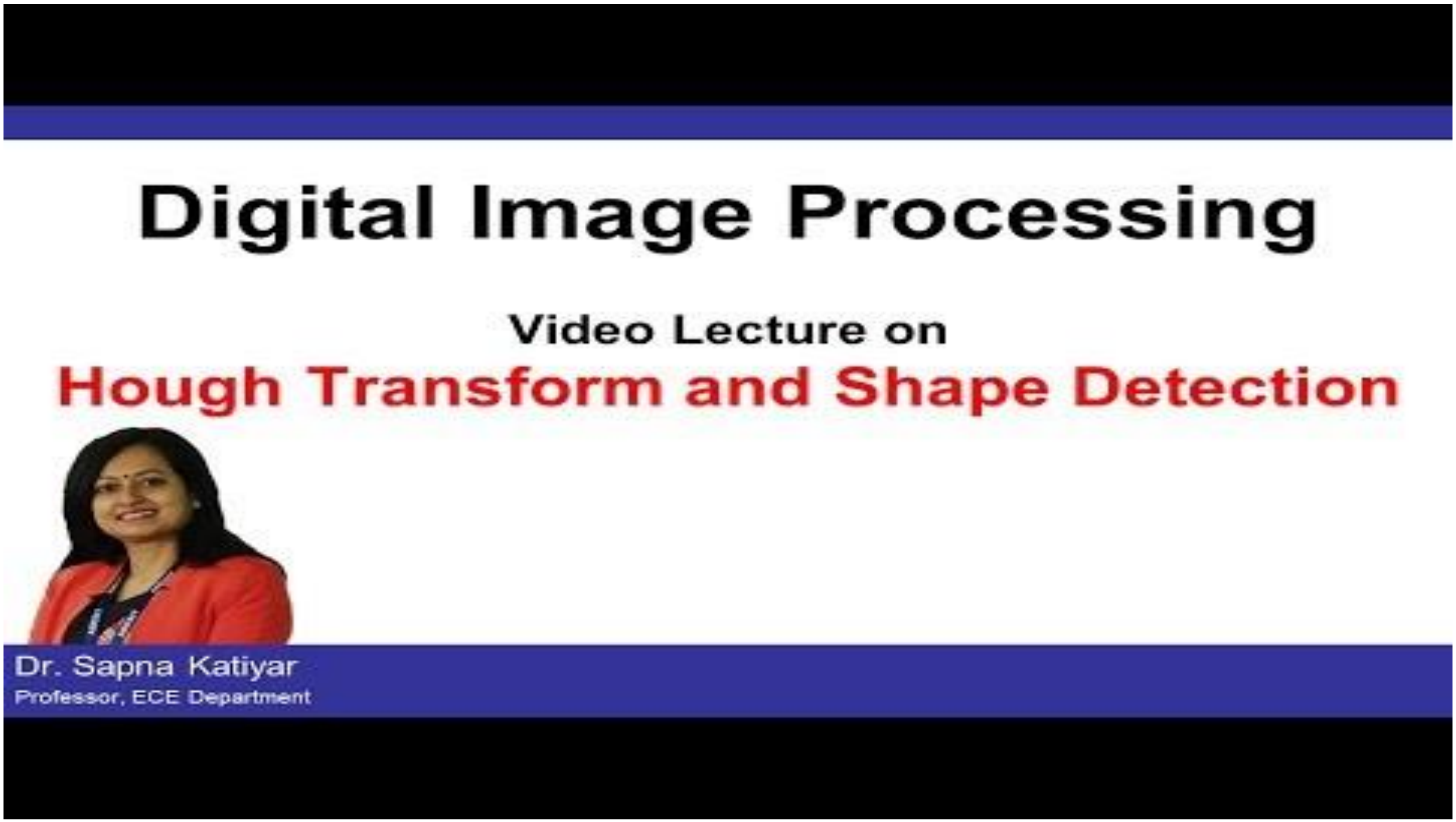


https://youtu.be/o-n7NoLArcs

Pseudo Code

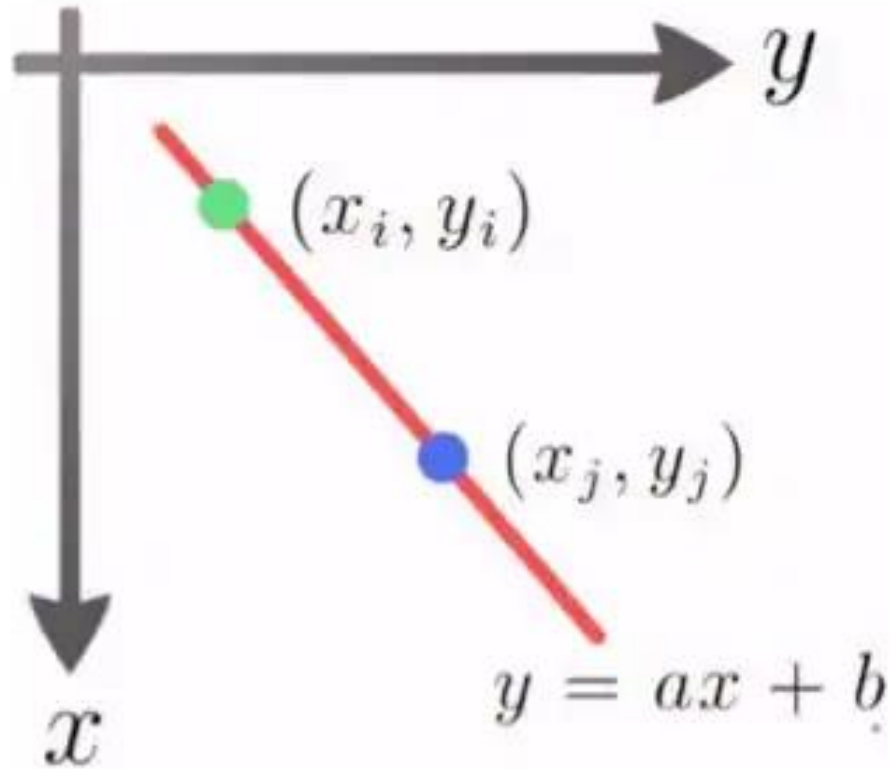
```
for all x
   for all y
      if edge point at (x,y)
         for all theta
            rho = x*cos(theta) + y*sin(theta)
            increment (add 1 to) the cell in H corresponding to (theta,rho)
         end
      end
   end
end
```

https://youtu.be/NFtPH2REs28



<u>Extra Slides</u>

Hough Transform



Haugh Transform

- Steps:
- Consider one valid edge point (xi,yi) in xy-plane & the equation of line passing through it can be,

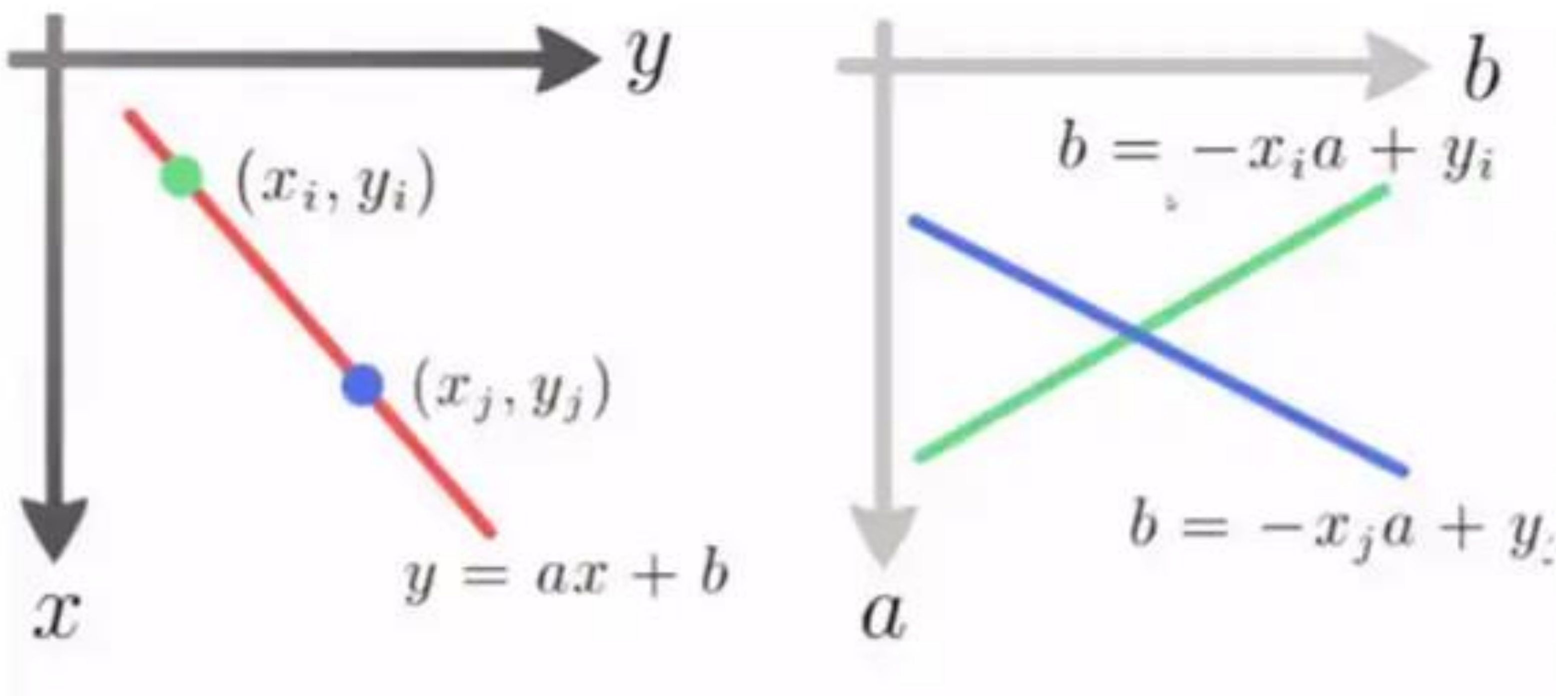
$$y_i = ax_i + b$$

- As it is a point, infinite lines will be passing through it given by above equation & different values of a & b.
- We can write this equation as,

$$b = -x_i a + y_i$$

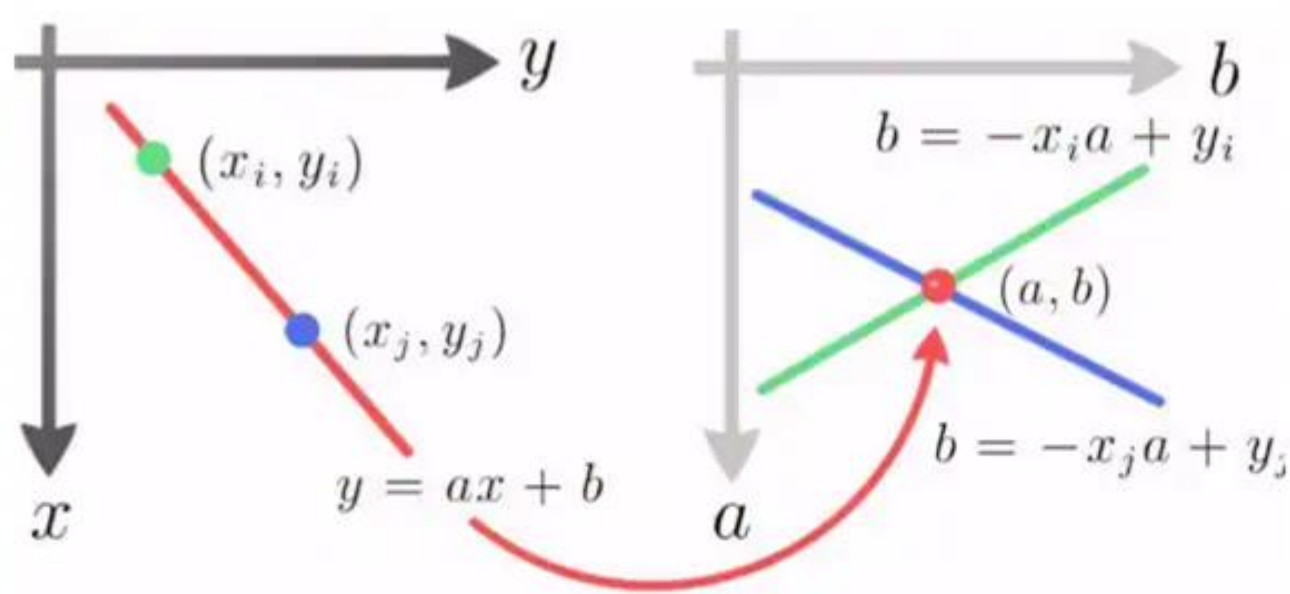
which gives us a line in ab-plane(parameter plane) passing through fixed pair (xi,yi).

Hough Transform



- Next, we will consider 2nd valid edge point (xj,yj) and find out equation in parameter plane. It will be,

Hough Transform



- Next, we will consider 2nd valid edge point (xj,yj) and find out equation in parameter plane. It will be,

$$b = -x_j a + y_j$$

- If these 2 points lies on a st line in xy-plane, then the two lines in parameter plane will intersect at point(a',b') where, a' is slope and b' is intercept of line passing through 2 points (xi,yi) and (xj,yj) in xy-plane.

Haugh Transform

- What is the drawback???
- Slope (a) will be infinite in case of vertical lines.
- Example:

  If 2 valid points are (3,1) and (3,2)

  Line eq in ab-plane will be,

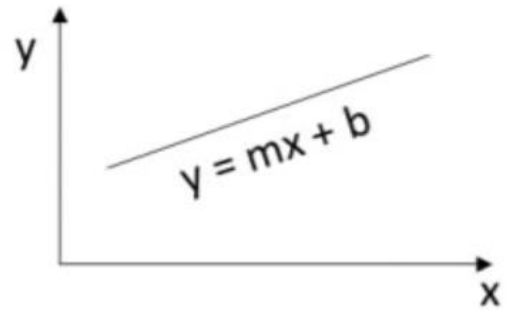
  (3,1)→b=-3a+1 & (3,2)→ b=-3a+2

  Here, slope of 2 lines is equal, hence they are parallel in ab-plane.

  We can not find point of intersection which gives us slope and intercept i.e a' and b' of line passing through (3,1) and (3,2) in xy-plane.
- Solution???

Hough Transform (continued)

- A line has two parameters (m,b)



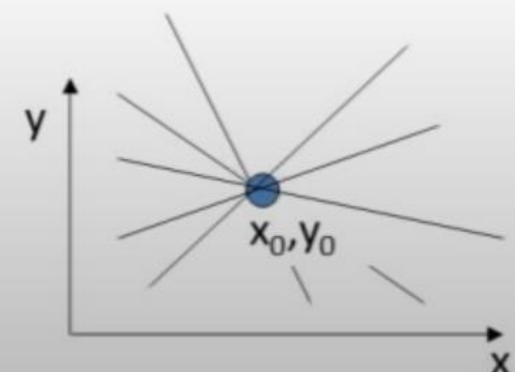
- Given a point (x_0,y_0), the lines that could pass through this point are all (m,b) satisfying

$$y_0 = m\, x_0 + b$$

- Or

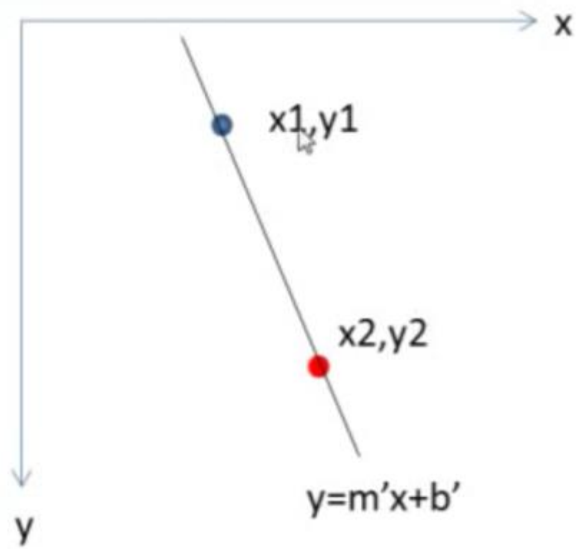
$$b = -x0\, m + y_0$$

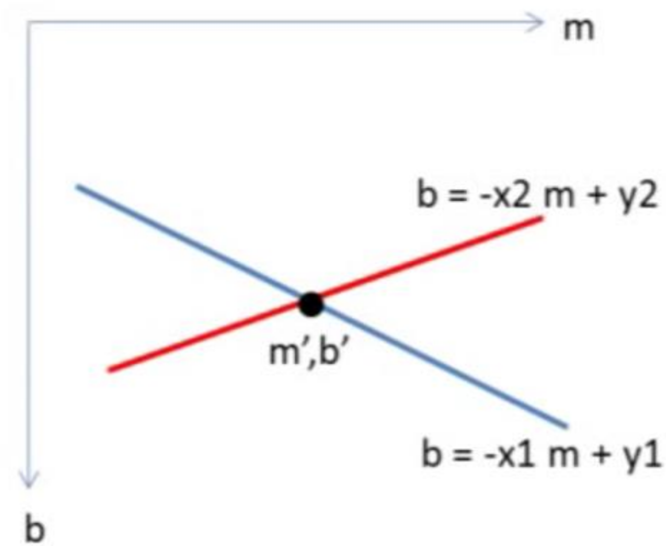
The equation b = -x0 m + y0 is a line in (m,b) space



Hough Transform (continued)

- All points on a line in image space, yield lines in parameter space which intersect at a common point
  - This point is the (m,b) of the line in image space





Hough Transform Algorithm

Load the image

Determine the image edges using any edge detector

Quantize the Parameter Space, P

Repeat the process for all pixels of image
If the pixel is an edge pixel, then

$$c = (-x)m + y$$

$$P(c.m) = P(c.m) + 1$$

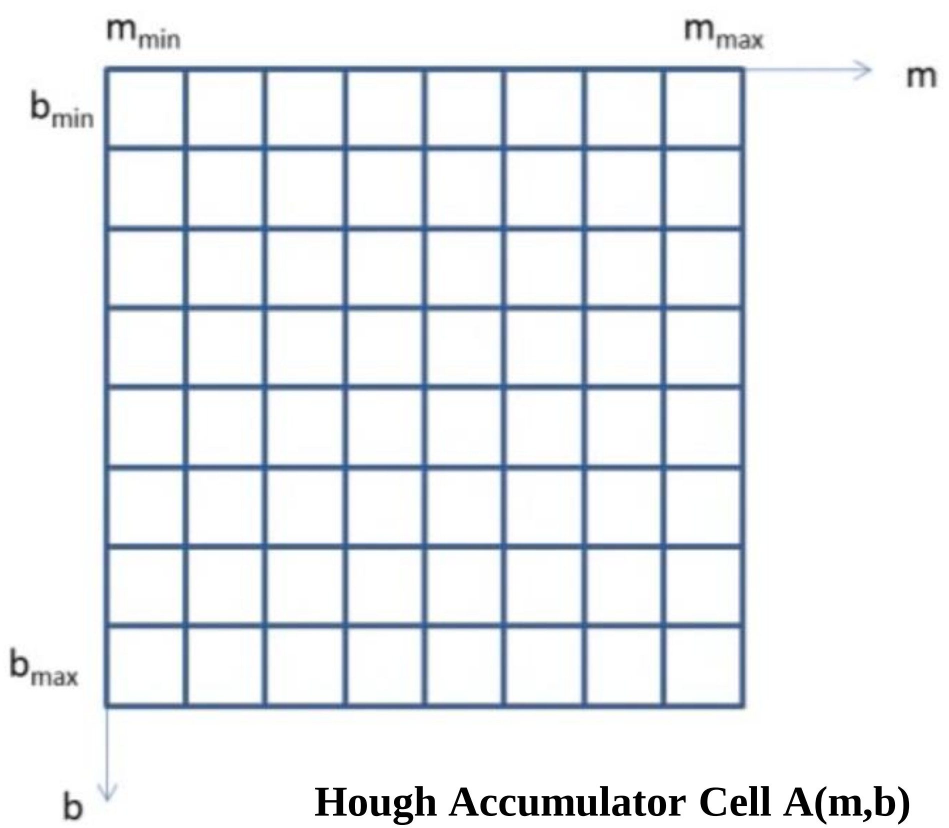
Show the Hough Space

Find the local maxima in parameter space

Draw the line using local maxima

Hough Transform Algorithm

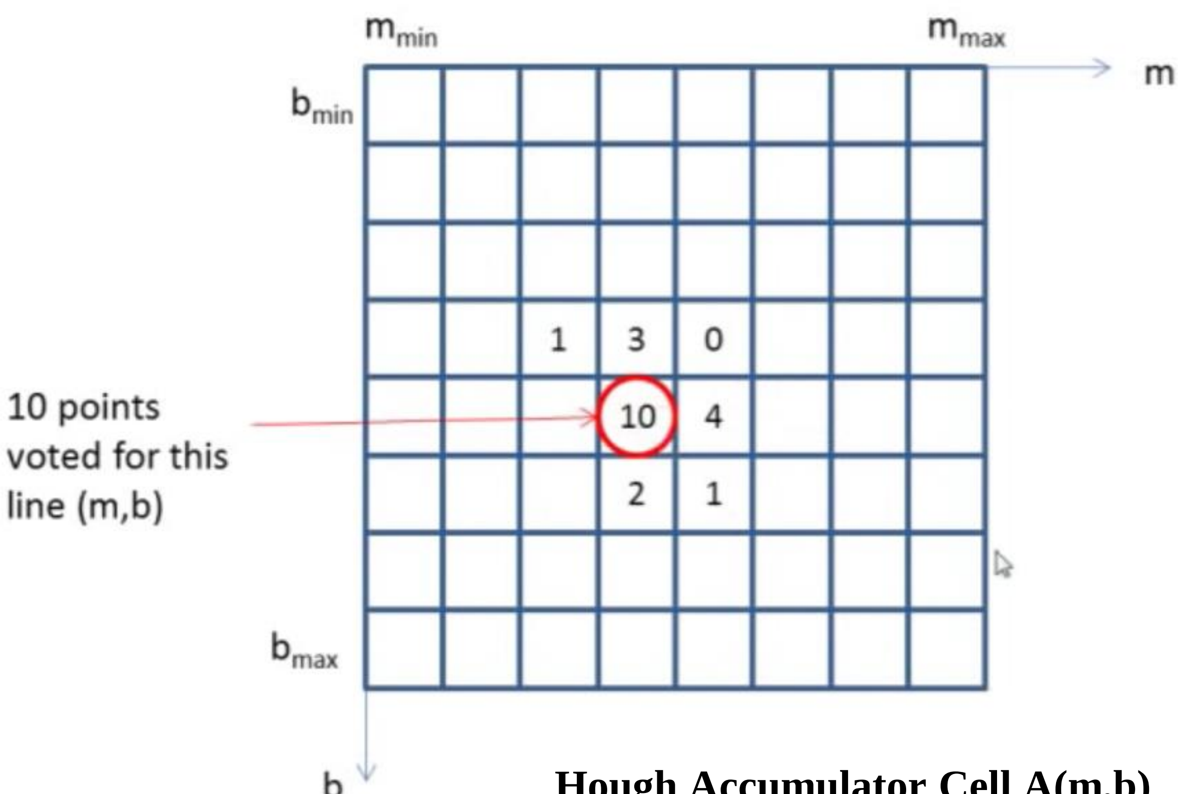
- Initialize an accumulator array A(m,b) to zero



Hough Accumulator Cell A(m,b)

Hough Transform Algorithm

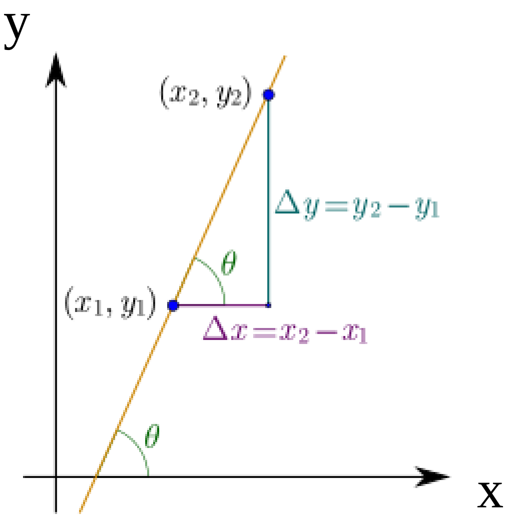
- Initialize an accumulator array A(m,b) to zero
- For each edge element (x,y), increment all cells that satisfy b = -x m + y
- Local maxima in A(m,b) correspond to lines



Hough Accumulator Cell A(m,b)

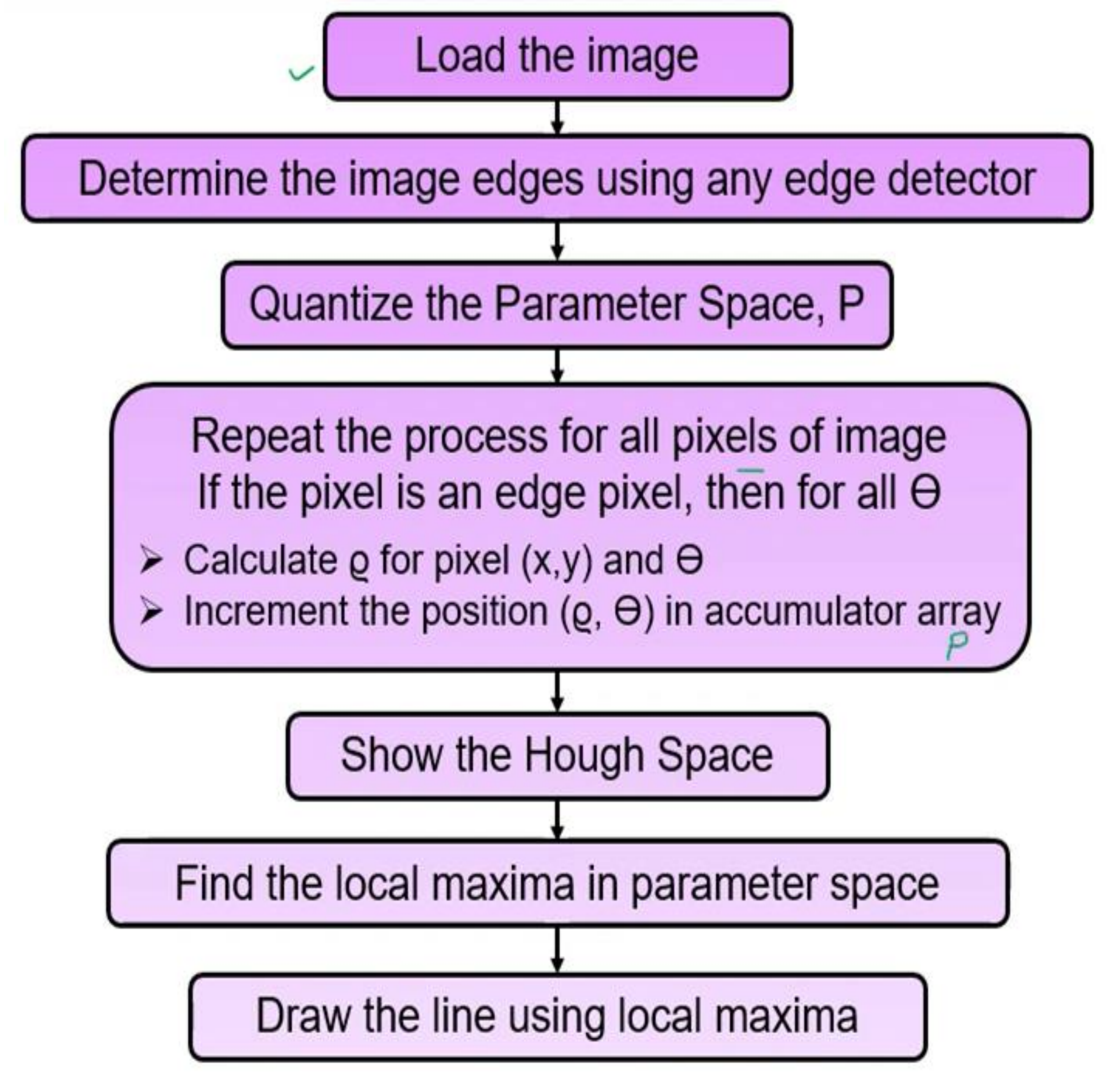
What is the slope of a line parallel to the y-axis?

- Parallel lines have the same slope.
- Vertical lines have an undefined slope.
- The y-axis is a vertical.
- A line that is parallel to the y-axis also has to be vertical.
- **The slope of a line parallel to the y-axis has a slope that is undefined**.



Slope = m = tanΘ

Hough Transform Algorithm in polar space



Better Parameterization

NOTE: $-\infty \leq m \leq \infty$

Large Accumulator

More memory and computations

Improvement: (Finite Accumulator Array Size)

Line equation: $\rho = -x\cos\theta + y\sin\theta$

Here $0 \leq \theta \leq 2\pi$

$0 \leq \rho \leq \rho_{\max}$

Given points (x_i, y_i) find (ρ, θ)

Hough Space Sinusoid